

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	44.0 / Female
Department	Gynaecology
Diagnosis	Acute cystitis
UTI Classification	Uncomplicated UTI
Sample Type	Pus Swab

Clinical Presentation

Chief Complaints: Dysuria;Frequency

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	38.0	%	20 - 40
Wbc	9100.0	cells/ μ L	4000 - 11000
Polymorphs	64.0	%	40 - 75
Crp	10.0	mg/L	< 10
Rft Serum Creatinine	0.9	mg/dL	0.6 - 1.3
Serum Uric Acid	4.3	mg/dL	3.5 - 7.2
Blood Urea	26.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Candida albicans
Bacteria Type	Fungus (Yeast; the most common cause of candiduria)

Antibiotic Susceptibility Profile

Predicted Sensitive	Amphotericin B, Voriconazole
Predicted Resistant	Fluconazole

Recommended Therapy

Voriconazole

Dosage: 200 mg orally every 12 hours for 7-14 days (adjust for hepatic function if needed)

Precautions: Monitor liver enzymes; avoid in patients with known hepatic impairment. No dose adjustment needed for normal renal function (creatinine 0.9 mg/dL). Check for drug interactions (e.g., CYP450 substrates).

Rationale: Voriconazole is listed as a sensitive agent against Candida albicans and is administered orally, making it convenient for an uncomplicated cystitis case. It avoids the nephrotoxicity associated with Amphotericin B.

Amphotericin B

Dosage: 0.3-0.5 mg/kg IV once daily (or bladder irrigation 50 mg in 500 mL saline, 1 L per day) for 7-10 days

Precautions: Nephrotoxic; monitor serum creatinine and electrolytes closely despite current normal renal function. Pre-hydrate with normal saline. Watch for infusion-related reactions and electrolyte disturbances (hypokalemia, hypomagnesemia).

Rationale: Amphotericin B is also sensitive for Candida albicans and can be used if oral therapy fails or the infection is severe. It provides broad fungicidal activity but requires careful renal monitoring.

Antibiotic Drug History

Clinical Summary

Infection: Uncomplicated acute cystitis (bladder infection) in a 44-year-old female.

Predicted pathogen: *Candida albicans* (fungal cystitis).

Resistance profile: Predicted resistance to fluconazole.

Sensitivity profile: Sensitive to voriconazole and amphotericin B.

Recommended therapy:

1. **Voriconazole** 200 mg PO q12 h for 7-14 days (dose adjustment if hepatic impairment).

- *Rationale:* Oral route is convenient for an uncomplicated case; voriconazole is a sensitive agent and avoids the nephrotoxicity of amphotericin B.

- *Precautions:* Monitor liver function tests; review concomitant CYP450 drugs; no renal dose adjustment needed (creatinine 0.9 mg/dL).

2. **Amphotericin B** 0.3-0.5 mg/kg IV daily (or bladder irrigation 50 mg in 500 mL saline once daily) for 7-10 days if oral therapy fails or infection is severe.

- *Rationale:* Provides fungicidal activity against *C. albicans* and serves as a second-line option.

- *Precautions:* Nephrotoxic-monitor serum creatinine, electrolytes (K⁺, Mg²⁺), and pre-hydrate with normal saline; watch for infusion reactions.

Key considerations: Ensure hepatic function is adequate before initiating voriconazole; maintain renal monitoring if amphotericin B is used. No dose modification is required for current normal renal function.